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Module 3: National Textbook Registry & Digital Passport (NTR)

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Audience	MINESEC/MINEDUB technical committees, bidding vendors, implementation engineers

This document replaces the narrative Chapter 19 specification with numbered, testable requirements, a normative data model, API contracts, and acceptance criteria. A vendor shall be able to bid against this document; an engineering team shall be able to build and be acceptance-tested against it.

Requirement keywords follow RFC 2119: SHALL (mandatory), SHOULD (recommended), MAY (optional). Every SHALL requirement has at least one acceptance criterion (AC) and is traceable in the Requirements Traceability Matrix (§13).

1. Scope

The National Textbook Registry (NTR) is the authoritative national register of every textbook title approved for use in Cameroonian schools, and — where policy dictates — a per-copy digital passport for physical textbook copies. It serves MINEDUB (primary) and MINESEC (secondary) under a single data model with ministry-scoped governance.

In scope: title registration and approval workflow; edition/version management; curriculum mapping; the National Textbook Identifier (NTID) scheme; per-copy passports with QR/barcode identifiers; lifecycle state tracking; condition history; replacement forecasting inputs; catalogue publication; offline-capable school transactions; APIs consumed by Procurement, Warehouse, Distribution, School Operations, Student Registry, and Analytics services.

Out of scope (owned by other modules): purchase orders and contracts (Procurement Service), physical warehouse operations (Nwidms), student identity (National Student Registry), digital e-book content hosting (NEDIH).

2. Definitions

Term	Definition
Title	An approved textbook work (e.g., "Mathematics for Form 1, 3rd Edition")
Edition	A versioned revision of a Title tied to a curriculum version
Copy	One physical printed book, individually identified
Batch	A group of copies from one print run, the minimum tracking unit where per-copy tracking is not cost-justified
Passport	The immutable event history attached to a Copy or Batch
NTID	National Textbook Identifier (title-level)
NCID	National Copy Identifier (copy-level)

3. Identifier Schemes (normative)

3.1 NTID — TITLE LEVEL

Format: `CM-TB-{MIN}-{SUBJ}-{GRADE}-{LANG}-{SEQ}-{ED}`

Segment	Values	Example
MIN	<code>B</code> (MINEDUB) / <code>S</code> (MINESEC)	S
SUBJ	3-letter subject code from national subject codeset (Annex A)	MAT
GRADE	<code>P1</code> – <code>P6</code> , <code>F1</code> – <code>F5</code> , <code>LS</code> / <code>US</code> (lower/upper sixth), <code>C1</code> – <code>C4</code> (CAP/technical)	F1
LANG	<code>EN</code> , <code>FR</code> , <code>BI</code>	EN
SEQ	4-digit sequence per (MIN,SUBJ,GRADE,LANG)	0007
ED	2-digit edition number	03

Example: `CM-TB-S-MAT-F1-EN-0007-03` .

- **FR-NTR-ID-01** The system SHALL generate NTIDs automatically on title approval; NTIDs SHALL be immutable and never reused, including for retired titles.
- **FR-NTR-ID-02** A new edition of an existing title SHALL retain all segments except `ED` , which increments.

3.2 NCID — COPY LEVEL

Format: `{NTID}-{BATCH:5}-{COPY:6}` encoded as a QR code (with human-readable fallback line) printed on a durable adhesive label or printed into the book cover at press.

- **FR-NTR-ID-03** NCIDs SHALL embed a check digit (ISO/IEC 7064 MOD 37-2) so that hand-typed entry of a damaged label can be validated offline.

- **FR-NTR-ID-04** QR payload SHALL be the bare NCID string (no URL), ≤ 64 characters, encodable at QR version 2 (25×25) for reliable scanning by low-end Android cameras.
- **FR-NTR-ID-05** Where ministry policy selects batch-level tracking for a title (policy flag on the Title record), the system SHALL track at batch granularity and SHALL NOT require per-copy scans for that title.

4. Data Model (normative)

Entity-relationship overview:

```

Title 1—N Edition 1—N PrintBatch 1—N Copy
Edition N—1 CurriculumVersion
Copy 1—N PassportEvent          (append-only)
Copy N—1 Location               (warehouse | school | in-transit | learner)
Copy 0..1—1 StudentAssignment  (active at most one)

```

4.1 TITLE

Field	Type	Req	Notes
ntid	string(32) PK	✓	§3.1, immutable
title_en / title_fr	string(300)	✓ (≥1)	bilingual
ministry	enum {MINEDUB, MINESEC}	✓	
subject_code	FK subject codeset	✓	Annex A
grade_code	enum	✓	§3.1
language	enum {EN, FR, BI}	✓	
publisher_id	FK Publisher Registry	✓	
isbn	string(17)	○	validated ISBN-13 when present
tracking_granularity	enum {COPY, BATCH}	✓	default BATCH; policy-set
approval_ref	string(100)	✓	ministerial decision number
approval_date	date	✓	
status	enum {DRAFT, APPROVED, SUSPENDED, RETIRED}	✓	state machine §5.1
expected_service_life_years	int (1–10)	✓	default 3
unit_cost_fcfa	decimal	✓	reference cost for planning
pages, weight_g, binding	int/int/enum	○	logistics planning

4.2 EDITION

edition_id PK, ntid FK, edition_no, curriculum_version_id FK, effective_academic_year, retirement_academic_year?, changes_summary

4.3 PRINTBATCH

batch_id PK, edition_id FK, procurement_contract_ref, printer_id FK Supplier Registry, quantity, print_date, qa_status enum{PENDING, PASSED, FAILED, PASSED_WITH_DEVIATION}, qa_report_ref

4.4 COPY

ncid PK, batch_id FK, lifecycle_state enum (§5.2), current_location_type enum{WAREHOUSE, TRANSIT, SCHOOL, LEARNER, DISPOSED}, current_location_id, condition enum{NEW, GOOD, FAIR, POOR, UNUSABLE}, condition_updated_at, school_year_received

4.5 PASSPORTEVENT (APPEND-ONLY, TAMPER-EVIDENT)

Field	Type	Notes
event_id	UUIDv7 PK	time-ordered
ncid / batch_id	FK (one required)	
event_type	enum (§5.2 transitions + CONDITION_ASSESSED, AUDITED, LOST_REPORTED, FOUND)	
occurred_at	timestampz	device time
recorded_at	timestampz	server time on sync
actor_user_id	FK IAM	
actor_role	string	denormalized for audit
location_type/location_id		
payload	jsonb	event-specific (condition, student_id, transfer ref...)
device_id	FK	offline provenance
prev_event_hash	sha256	hash chain per copy (FR-NTR-AUD-02)

- **FR-NTR-DM-01** PassportEvent SHALL be append-only: no UPDATE or DELETE grants exist for any application role; corrections are made by compensating events (event_type=CORRECTION , payload referencing the corrected event).
- **FR-NTR-DM-02** Each PassportEvent SHALL store the SHA-256 of the previous event for the same copy, forming a verifiable hash chain. A nightly job SHALL verify chains and raise an audit alert on breakage.
- **FR-NTR-DM-03** All entities SHALL carry created_at, created_by, updated_at, updated_by, sync_origin (CENTRAL | device_id).

5. State Machines (normative)

5.1 TITLE STATUS

`DRAFT` → `APPROVED` → {`SUSPENDED` ⇌ `APPROVED`} → `RETIRED`. `RETIRED` is terminal. Only users holding `curriculum.approve` may execute `DRAFT→APPROVED` and `→RETIRED`; both require a ministerial reference string.

5.2 COPY LIFECYCLE_STATE

```
PRINTED → IN_WAREHOUSE → IN_TRANSIT → AT_SCHOOL → ASSIGNED → RETURNED(→AT_SCHOOL)
AT_SCHOOL|ASSIGNED → UNDER_REPAIR → AT_SCHOOL
any state → LOST → (FOUND → previous state)
AT_SCHOOL|IN_WAREHOUSE → RETIRED → DISPOSED
```

- **FR-NTR-SM-01** The system SHALL reject any event implying an illegal transition (HTTP 409 with machine-readable code `ILLEGAL_TRANSITION`), except events arriving via offline sync, which SHALL be quarantined for reconciliation (§9.4) rather than rejected.
- **FR-NTR-SM-02** `ASSIGNED` SHALL require a valid `student_id` resolvable in the National Student Registry (or a locally cached student record when offline).

6. Functional Requirements

6.1 TITLE & EDITION MANAGEMENT

ID	Requirement (SHALL)	Acceptance criterion
FR-NTR-01	Register a textbook title with all mandatory §4.1 fields; incomplete submissions are rejected with field-level errors	POST with a missing mandatory field returns 422 listing exactly the missing fields; complete POST returns 201 with generated NTID
FR-NTR-02	Enforce the title approval workflow: creation in DRAFT; transition to APPROVED only by <code>curriculum.approve</code> role with <code>approval_ref</code>	A <code>curriculum.edit</code> user attempting approval receives 403; approval without <code>approval_ref</code> returns 422
FR-NTR-03	Create a new Edition linked to a CurriculumVersion; on the effective academic year, prior editions flagged <code>superseded</code>	Catalogue query for AY N returns only editions effective in N; superseded edition appears with <code>superseded=true</code>
FR-NTR-04	Prevent procurement referencing RETIRED or SUSPENDED titles (validation API consumed by Procurement Service)	Validation call for a RETIRED NTID returns <code>procurable=false</code> with reason code
FR-NTR-05	Retire a title only when a retirement plan exists (successor NTID or explicit “no successor” declaration)	Retirement without plan payload returns 422

6.2 PRINT BATCHES & QA

ID	Requirement	Acceptance criterion
FR-NTR-06	Register print batches against an edition with contract reference and quantity; generate NCIDs (or batch ID) at registration; export label print file (CSV + PDF)	Registering a 5,000-copy batch generates exactly 5,000 unique valid NCIDs in < 30 s; label export validates against §3.2
FR-NTR-07	Record batch QA outcome; a FAILED batch's copies cannot enter IN_WAREHOUSE	Warehouse receipt scan against FAILED batch returns <code>QA_BLOCKED</code>

6.3 PASSPORT & MOVEMENT

ID	Requirement	Acceptance criterion
FR-NTR-08	Record warehouse receipt, dispatch, school receipt, student assignment, return, repair, loss, retirement as PassportEvents via scan or manual NCID entry	Each event type creates exactly one PassportEvent; copy state and location update atomically with the event
FR-NTR-09	Show any copy's full passport (all events, chronological, with actor and location) to authorized users in ≤ 2 s for a copy with ≤ 500 events	Timed test on seeded copy passes at p95
FR-NTR-10	Support bulk operations: one dispatch/receipt action covering up to 10,000 copies (by batch or scan-list)	Bulk receipt of 10,000 copies completes ≤ 60 s and creates per-copy events
FR-NTR-11	Record condition on every return and on annual verification, with photo attachment optional (≤ 2 MB, compressed client-side)	Return without condition value is rejected client-side; condition history query returns time series
FR-NTR-12	Annual stock verification campaign: ministry opens a campaign window; schools submit verification scans; system computes per-school reconciliation (expected vs scanned vs missing)	Campaign report lists every school with counts; a school with 100 expected and 97 scanned shows 3 in <code>unverified</code>

6.4 CATALOGUE & REPORTING

ID	Requirement	Acceptance criterion
FR-NTR-13	Publish a public, read-only approved-textbook catalogue (web + JSON API), filterable by ministry, subject, grade, language, academic year	Unauthenticated GET returns approved titles only; DRAFT/SUSPENDED titles never appear
FR-NTR-14	Provide the standard reports of Annex B (edition history, lifecycle distribution, condition analysis, replacement forecast input, coverage by school/division/region) as on-screen views and CSV/XLSX export	Each report renders with region→division→school drill-down; export matches on-screen data
FR-NTR-15	Compute textbook-per-learner coverage per school per title, joining School Registry enrolment with AT_SCHOOL+ASSIGNED copy counts	For seeded school (400 learners, 320 assigned copies of a title) coverage shows 0.80
FR-NTR-16	Expose replacement-forecast inputs per title per school year: copies by condition band, age distribution, historical loss rate	API returns the documented JSON structure; Analytics Service integration test passes

6.5 ROLES & PERMISSIONS

- **FR-NTR-17** The module SHALL enforce the permission matrix below via the central IAM service (OIDC; roles as claims). Every mutation SHALL be attributable to a named user — no shared accounts.

Action	Curriculum Officer	Procurement Officer	Warehouse Officer	Head Teacher / School Store Manager	Teacher	Inspector	Read-only National
Create/edit DRAFT title	✓						
Approve/retire title	✓ (approve right)						
Register batch / QA		✓	✓ (QA)				
Warehouse receipt/dispatch			✓				
School receipt / return / condition				✓	✓ (assign/return own class)		
Assign to student				✓	✓		
Verification campaign submit				✓		✓ (spot check)	
View any passport	✓	✓	✓ (own warehouse)	✓ (own school)	own class	✓	✓

- **FR-NTR-18** School-scoped roles SHALL only read/write data for their own school (row-level enforcement server-side, not UI-only).

7. API Contracts (normative, summary)

Base path `/api/v1/ntr`. JSON, UTF-8, OAuth2 bearer (client-credentials for service-to-service, auth-code for users). Errors use RFC 7807 problem+json. Full OpenAPI 3.1 file is deliverable D-NTR-API (generated from this section; this section governs on conflict).

Endpoint	Method	Purpose
/titles	GET, POST	search/register titles
/titles/{ntid}	GET, PATCH	detail / edit DRAFT fields
/titles/{ntid}:approve, :suspend, :retire	POST	state transitions
/titles/{ntid}/editions	GET, POST	editions
/titles/{ntid}/procurability	GET	validation for Procurement (FR-NTR-04)
/batches	POST	register print batch, returns NCID range + label export URL
/batches/{id}/qa	POST	QA outcome
/copies/{ncid}	GET	copy + current state
/copies/{ncid}/passport	GET	full event history
/events:bulk	POST	up to 10,000 events per call (dispatch, receipt, assignment...) idempotent by client-supplied <code>event_uuid</code>
/catalogue	GET	public approved catalogue
/reports/{report_code}	GET	Annex B reports, `?format=json`
/campaigns	POST, GET	verification campaigns
/sync/pull, /sync/push	POST	offline device sync (§9)

- **FR-NTR-API-01** All list endpoints SHALL support cursor pagination (`limit ≤ 500` , `next_cursor`) and field filtering; `/events:bulk` SHALL be idempotent — replaying the same `event_uuid` set returns the original result and creates no duplicates.
- **FR-NTR-API-02** Breaking changes SHALL only ship under a new `/api/v2` path; `/api/v1` supported ≥ 24 months after v2 GA.

8. Non-Functional Requirements

ID	Category	Requirement
NFR-NTR-01	Scale	Support $\geq 30,000,000$ Copy records, $\geq 300,000,000$ PassportEvents, $\geq 25,000$ school tenants without architecture change (partition PassportEvent by school-year)
NFR-NTR-02	Performance	p95 API latency ≤ 500 ms for single-entity reads, ≤ 3 s for reports at division level, on reference infrastructure defined in the Enterprise Architecture volume
NFR-NTR-03	Throughput	Sustain 200 sync sessions/minute during back-to-school peak (Sept), each pushing $\leq 5,000$ events
NFR-NTR-04	Availability	99.5% monthly for central services; school operations unaffected by central outage (offline-first)
NFR-NTR-05	Offline	Full school workflow (receipt, assignment, return, condition, verification) executable with zero connectivity for ≥ 90 consecutive days on an Android 10+ device with 2 GB RAM
NFR-NTR-06	Data protection	Compliant with Cameroon Law N° 2010/012 (cybersecurity/cybercrime) and the platform Data Governance Framework; student-copy links are personal data: encrypted at rest (AES-256), access-logged
NFR-NTR-07	Audit	100% of mutations attributable to user + device + timestamp; passport hash chains verified nightly (FR-NTR-DM-02)
NFR-NTR-08	Localization	Full UI in French and English; all user-facing enumerations bilingual
NFR-NTR-09	Portability	No proprietary cloud service without an open-standard exit path; database PostgreSQL ≥ 15 ; deployable on national data centre or IaaS
NFR-NTR-10	Accessibility & inclusion	Mobile app screen-reader (TalkBack) compatible; touch targets ≥ 48 dp; WCAG 2.1 AA contrast; every critical workflow completable via icon + short-label path without reading prose (low-literacy usability); coverage reports disaggregable by school characteristics (EDUOS-ESS-001 INC-01/02) — applies equally to NSR and NWIDMS UIs

9. Offline Synchronization Profile

Applies the National Offline Synchronization Engine to NTR data:

- 9.1 Device enrolment.** School devices enrol via IAM; each receives a device certificate and a school-scoped data partition (its own copies, students, titles catalogue).

2. **9.2 Pull.** `sync/pull` delivers catalogue + school partition as delta since last sync cursor (compressed; initial seed ≤ 50 MB for a 2,000-learner school).
 3. **9.3 Push.** Events created offline carry `event_uuid` (client UUIDv7), device time, and device ID; push is chunked and resumable.
 4. **9.4 Conflict rules (normative).** (a) PassportEvents never conflict — they append and are re-ordered by `occurred_at` (b) an event implying an illegal transition after reordering is quarantined with status `NEEDS_RECONCILIATION` and surfaced to the division office work-queue, never silently dropped (c) copy `condition` resolves last-writer-wins by `occurred_at` (d) student assignment conflicts (two schools claim one copy) auto-open a reconciliation case.
- **FR-NTR-SYNC-01** Quarantined events SHALL be resolvable (accept / correct / reject) by `division.reconcile` role, with resolution recorded as a PassportEvent.

10. UI Requirements (summary)

Mobile-first Android app + responsive web, per the EduOS Cameroon Design System Specification. Mandatory screens: scan-first “receive stock”, “assign to learner” (scan copy → scan/select learner), “return & condition”, “verification campaign”, school dashboard (coverage, outstanding returns). Scan-to-confirmation SHALL take ≤ 3 taps. Every screen SHALL function offline with a visible sync-status indicator.

11. Migration & Data Seeding

- **FR-NTR-MIG-01** Provide a bulk import tool (CSV templates, Annex C) for the existing approved-textbook list (est. 2,000–4,000 titles) and school stock declarations, with a validation report (accepted/rejected rows with reasons) before commit.
- **FR-NTR-MIG-02** Support “brownfield” copy registration: schools may register pre-existing unlabelled stock as batch-level quantities per title, upgradeable to per-copy tracking when labels are applied.

12. Acceptance & Test Strategy

Acceptance = 100% of SHALL requirements pass their ACs in a witnessed User Acceptance Test on the pilot environment, plus: a 10-school pilot operating 60 days with $\geq 95\%$ of textbook movements captured digitally; one full offline cycle (30 days disconnected, then successful sync with zero data loss); load test at NFR-NTR-03 rates. The vendor SHALL deliver automated test suites (API contract tests + E2E) that the Ministry can re-run.

13. Requirements Traceability Matrix

Requirement range	Verification method	Test deliverable
FR-NTR-ID-01...05	Unit + contract test	TST-NTR-ID
FR-NTR-DM-01...03	DB privilege audit + chain-verification test	TST-NTR-DM
FR-NTR-SM-01...02	State-machine property tests	TST-NTR-SM
FR-NTR-01...18	API contract + E2E UAT scripts	TST-NTR-FN
FR-NTR-API-01...02	Contract tests (Schemathesis or equivalent)	TST-NTR-API
NFR-NTR-01...09	Load test, offline soak test, security audit	TST-NTR-NFR
FR-NTR-SYNC-01, MIG-01...02	Pilot scenario tests	TST-NTR-PIL

Annex A — Subject codeset (extract)

MAT Mathematics · ENG English Language · FRE French · PHY Physics · CHE Chemistry · BIO Biology · HIS History · GEO Geography · CIV Citizenship · ECO Economics · CSC Computer Science · LIT Literature · SCI Integrated Science (primary) · ... *(full codeset maintained as reference data by the Curriculum Service; initial load in deliverable D-NTR-REF)*

Annex B — Standard report codes

RPT-CAT approved catalogue · RPT-EDH edition history · RPT-LCS lifecycle status distribution · RPT-CND condition analysis · RPT-COV coverage by school/division/region · RPT-RPL replacement forecast input · RPT-VER verification campaign results · RPT-LOSS loss analysis.

Annex C — Import templates

titles.csv (columns = §4.1 mandatory fields), school_stock.csv (school_id, ntid, quantity, condition_band, year_received). Templates with validation rules ship as deliverable D-NTR-IMP.